

STANDARD OPERATING PROCEDURE

Plate-Reader Redox Phenotyping

Microplate protocols for H₂O₂, lipid peroxidation, antioxidant enzymes, glutathione, ascorbate and proline in maize and wheat

SOP ID / version	SOP-RDX-02 · v2.0 (29 September 2026) · companion to SOP-RDX-01
Species	<i>Zea mays</i> L. (maize) and <i>Triticum aestivum</i> L. (wheat); leaves and roots
Status	Research-use protocol. Conditions marked as pilot must be validated for the local tissue, instrument and treatment matrix before large-scale use. Acceptance criteria are proposed laboratory criteria, not universal limits from the cited papers.
Prepared / reviewed / approved	

How this document is organised. **Part A** sets the sampling, optical and validation requirements that apply to every assay. **Part B** contains the twelve assay procedures (M1–M12). **Part C** states what cannot be moved onto a plate reader. **Part D** covers plate capacity, timing, work allocation and the general equations. **Part E** records the evidence base and the validation still outstanding.

Read Part A before running anything. The optical-path and extraction conventions there determine whether the numbers from Part B mean anything.

Instrument requirements at a glance

Confirm these capabilities before starting. An assay whose wavelength your reader cannot transmit, or whose plate type you do not have, cannot be rescued by data processing.

Assay	Mode and wavelength	Plate
M1 H ₂ O ₂ , Amplex Red · M3 root peroxide	Fluorescence: excitation 530–560 nm, emission near 590 nm; fix gain within the study	Black, flat bottom
M7 catalase (Route A)	Kinetic absorbance, 240 nm	UV-transparent, with verified UV transmission
M8 APX	Kinetic absorbance, 290 nm	UV-transparent, with verified UV transmission
M9 GR (Route A)	Kinetic absorbance, 340 nm	Check actual plate transmission; UV-transparent is the conservative choice
M2 H ₂ O ₂ , potassium iodide	Absorbance, 350 nm	Check transmission and blank absorbance at 350 nm
M4 TBARS · M5 protein · M6 SOD · M10 glutathione · M11 ascorbate · M12 proline	Absorbance at the stated wavelengths, approximately 412–600 nm	Clear, compatible with the reaction solvents, acids and fill volume

Two things the reader does not do. **Sample preparation happens off the instrument.** Extraction, centrifugation and the 95 °C reactions in M4 and M12 are bench steps. Heat those reactions in solvent- and temperature-rated screw-cap tubes or validated sealed polypropylene vessels, then transfer to a reading plate. Never heat an ordinary assay plate to 95 °C.

96 wells is not 96 samples. Blanks, standards, paired controls, QC and any technical duplicates occupy wells before a single sample does. Part D1 reserves 24 of 96 wells for these, leaving 72 sample wells, and gives the resulting plate counts per assay. M1 fits 16 samples per plate; M4 fits 32.

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A Sampling and common requirements

A1 Biological design and interpretation

Define the experimental unit before sampling: a pot, an independently treated plant or a deliberately pooled biological replicate. Technical wells do not increase biological replication. Balance treatments across harvest order, extraction batches and plates. Record species, genotype, organ, leaf rank or root zone, developmental stage, treatment, time since stress onset or drainage, and clock time.

For roots, standardize washing and blotting and minimize handling before freezing; document the delay. Freeze tissue promptly in liquid nitrogen and store separate aliquots at -80°C . Do not assume frozen storage preserves every analyte indefinitely. Establish the acceptable storage interval and freeze–thaw tolerance using representative material. M3 requires living tissue and a separate harvest.

Higher antioxidant enzyme activity or proline is not, by itself, evidence of greater resilience. Interpret these alongside damage, growth and recovery measurements. Bulk extracts mix cellular compartments and cannot establish *in vivo* redox potential or compartment-specific oxidative stress. Fresh-weight normalization is affected by tissue hydration; collect dry-matter fraction on a separate matched aliquot where relevant.

A2 Optical path and units

For absorbance, $A = \epsilon cl$. The estimate $l = V/A_{\text{well}}$ gives about 0.625 cm at 200 μL in a well with 0.32 cm^2 cross-sectional area, but this is not a calibrated optical path. Use either a same-matrix standard curve or a validated path-length measurement. A calibrated extinction-coefficient method is acceptable for publication. A geometric estimate alone is insufficient for accurate absolute activities.

Water-based near-infrared path correction can fail in solvent-rich, turbid or strongly absorbing mixtures. Verify compatibility with the instrument manual. If absorbance has already been normalized to 1 cm, use $l = 1$ cm in the calculation. Do not correct twice. Standard curves and path correction can coexist if standards and samples receive identical treatment, but a curve normally makes a separate path correction unnecessary.

Measurement	Plate and instrument requirement
240 and 290 nm	UV-transparent plates and verified UV transmission; CAT and APX respectively.
340 and 350 nm	Check the actual plate transmission and blank absorbance. UV-transparent plates are a conservative choice; ordinary plates are not universally unusable at 340 nm.
412 to 600 nm	Clear plates compatible with the reaction solvents, acids and fill volume.
Amplex Red fluorescence	Black plates; excitation 530–560 nm and emission near 590 nm. Fix gain within the study or use a validated cross-run calibration.
Heating	Use solvent- and temperature-rated screw-cap tubes or validated sealed polypropylene vessels. Never heat ordinary assay plates to 95°C .

A3 Extraction for soluble enzyme activity

The original document depended on an unavailable P8 extraction method. The following is a proposed starting extraction for pilot validation, not a verified replacement for every isoenzyme: extract 100 mg frozen powder with 1.00 mL ice-cold 50 mM potassium phosphate, pH 7.0, containing 1 mM EDTA and 1% insoluble PVPP. Homogenize cold, centrifuge at $16,000 \times g$ for 10 min at 4°C , and transfer the clear supernatant. For APX, use a separate aliquot with freshly added 2 mM ascorbate. Assay immediately in small batches.

This extraction targets soluble activity. Membrane-bound APX or other insoluble fractions are not quantitatively recovered. Detergent may be introduced only after assay-compatibility and recovery tests. Do not add DTT or β -mercaptoethanol to the common extract; these can interfere with redox assays. PVPP-

containing enzyme extracts must not be reused for M1. Measure M5 protein on each actual enzyme extract, including the separate APX extract.

Record Vext, the defined extraction volume, and mass m for every sample. Where content is calculated using added solvent volume, report that convention as an operational extraction estimate. Tissue water can matter at low solvent-to-tissue ratios, especially M1; measure/validate the effective liquid volume or demonstrate negligible impact. Do not substitute recovered supernatant volume without accounting for analyte retained in the pellet.

A4 Validation before the main experiment

Pilot each species × organ and representative treatment matrix. Test at least three extract dilutions, a concentration range spanning the samples, extraction and post-extraction spikes for metabolites, reagent blanks, and within-day stability. For enzymes, demonstrate proportionality to extract amount and an initial linear interval with little substrate depletion. Check at least five usable time points where feasible.

Suggested internal starting criteria are 80–120% spike recovery, $\leq 10\%$ technical CV above the quantification limit and $\leq 15\%$ interplate QC CV. These are proposed laboratory criteria, not universal acceptance limits from the cited papers. Investigate failures and revise the assay before accepting results. Near zero, use an absolute precision limit instead of CV. Establish an LOQ from low-level precision and bias; a high calibration R^2 alone is insufficient.

Do not automatically divide concentrations by a treatment-specific spike-recovery fraction. Recovery failure may reflect nonlinear inhibition, analyte formation or degradation. Use recovery correction only if its validity and uncertainty have been demonstrated in advance. Keep raw and corrected results traceable.

B Assay procedures

M1 Tissue hydrogen peroxide by Amplex Red

Basis Brumbarova et al. (2016), Chakraborty et al. (2016), the Amplex Red manufacturer instructions and Votyakova and Reynolds (2004). The paired catalase workflow below is an adaptation. Report catalase-sensitive H_2O_2 equivalents recovered under the defined extraction and incubation conditions.

HRP-dependent Amplex Red oxidation produces fluorescent resorufin. Catalase suppression supports H_2O_2 involvement but does not prove that peroxide was present in intact tissue. H_2O_2 generated after extraction also contributes to the untreated signal and can be removed in the catalase-treated aliquot; it therefore remains in the difference. Spike recovery and dilution tests do not fully resolve this origin problem.

Reagents and calibration

Extract in ice-cold 20 mM potassium phosphate, pH 6.5. Use a freshly prepared 2× detection solution containing 100 μM Amplex Red and 0.2 U mL^{-1} HRP. Prepare the detection solution in the manufacturer reaction buffer, 50 mM sodium phosphate pH 7.4, rather than silently replacing it with extraction buffer. The final mixed pH must be checked in representative matrix. Prepare 10 mM Amplex Red in DMSO and HRP stock according to supplier storage instructions; protect from light and repeated thawing.

Catalase stock: approximately 1,000 U mL^{-1} in extraction buffer. Confirm that the chosen treatment removes a known H_2O_2 spike in the matrix. Prepare fresh H_2O_2 working stocks from a standardized concentrated solution. Verify a suitably concentrated dilution at A240 in a quartz cuvette; the low-micromolar standards are too dilute for accurate A240 standardization.

Calibration x-axis: nominal H_2O_2 concentration in the untreated 50 μL reaction. Use 0, 0.5, 1, 2, 3, 4 and 5 μM . To allow both the 66/60 pretreatment dilution and the 25/50 reaction dilution, prepare starting standards at 2.2× these concentrations in extraction buffer. Process each through the same A/B pretreatment as samples. Catalase wells are labelled by the concentration before catalase removal.

Procedure

1. Resuspend 30 mg frozen powder in 200 μL extraction buffer. Record mass and extraction time. Centrifuge 3 min at $16,200 \times g$ at 4 °C and transfer supernatant onto ice. Start with batches of 8–12 until stability is established.
2. Split 60 μL into tube A and 60 μL into tube B. Add 6 μL buffer to A and 6 μL catalase stock to B. Incubate both for exactly 10 min at room temperature. This extra incubation is an unvalidated loss/formation window; test it in the pilot.
3. For each aliquot set up one reaction with 25 μL aliquot plus 25 μL detection solution, and one matched no-probe background with 25 μL aliquot plus 25 μL HRP-containing buffer without Amplex Red. Match DMSO in no-probe wells. A single full control set therefore uses four wells per sample. For duplicates of all four wells, prepare larger pretreatment aliquots at the same 10:1 ratio; 60 + 6 μL is insufficient.
4. Incubate 30 min in the dark, with controlled addition-to-read time. Read at the fixed fluorescence settings. Verify in a pilot time course that samples and standards behave comparably; dilute high signals before repeating.

Calculation and controls

$S = (\text{FA,probe} - \text{FA,no-probe}) - (\text{FB,probe} - \text{FB,no-probe})$. Apply the same expression to every standard and reagent control. Fit $S = \text{intercept} + \text{slope} \times C_{\text{well}}$ within the validated range; do not force the intercept to zero without justification. $C_{\text{well}} = (S_{\text{sample}} - \text{intercept})/\text{slope}$.

$C_{\text{ext}} (\mu\text{M}) = C_{\text{well}} \times 2 \times 1.1 \times D$, where D is any additional extract dilution before pretreatment. Content ($\text{nmol g}^{-1} \text{FW}$) = $C_{\text{ext}} \times V_{\text{ext}} (\text{mL})/m (\text{g})$. For $C_{\text{well}} = 2 \mu\text{M}$, $D = 1$, $V_{\text{ext}} = 0.200 \text{ mL}$ and $m = 0.030 \text{ g}$, the solvent-volume estimate is 29.3 nmol g^{-1} . The effective-volume caveat in A3 is material here.

Include reagent-only controls, the processed standard series, matrix spikes and a pooled QC. Excess peroxide can overoxidize resorufin to nonfluorescent resazurin. Reductants can generate peroxide or

suppress detection. Do not add acid, PVPP or charcoal to this extraction without separate validation. A no-HRP pilot can reveal endogenous peroxidase effects. If treatment matrices fail validation, use an independently validated method for confirmation.

M2 Hydrogen peroxide screening by potassium iodide

Basis Junglee et al. (2014). The one-step method is a peroxide-equivalent screen; background subtraction does not make iodide oxidation H_2O_2 -specific. The paper optimized tomato tissues, not maize and wheat.

1. For each sample prepare 1.00 mL reagent: 250 μL 0.1% TCA, 500 μL 1 M KI and 250 μL 10 mM potassium phosphate. Final concentrations before tissue addition are 0.025% TCA, 0.5 M KI and 2.5 mM phosphate. Prepare a matched no-KI reagent by replacing KI with water. Pilot phosphate stock pH; pH 5.8 was effective for tomato leaves and is not a universal final reaction pH.
2. Extract matched 150 mg portions of the same frozen powder in the KI and no-KI reagents. Record both masses and keep the same mass-to-volume ratio. Incubate 10 min at 4 °C in the dark, then centrifuge 15 min at 12,000 $\times$ g at 4 °C.
3. Transfer 200 μL to UV-transparent wells, incubate 20 min at 20–22 °C in the dark and read A350. Record timing consistently. A390 is a separately calibrated alternative, not an interchangeable readout.
4. Prepare H_2O_2 standards through the complete KI reagent chemistry, including identical acid, salt, phosphate and incubation conditions. A stock in TCA alone is not a reacted calibration standard. Define the x-axis as final concentration in the 1 mL extraction/reaction mixture.

Signal = (A_{sample,KI} – A_{sample,no-KI}) – (A_{reagent,KI} – A_{reagent,no-KI}). Read C_{mix} in μM from the identically blanked calibration. Content (nmol g^{-1} FW) = C_{mix} $\times$ V_{mix} (mL) $\times$ D/m; divide by 1,000 for $\mu\text{mol g}^{-1}$. No extra factor arises from transferring 200 μL to the reading plate. Validate extraction recovery, pigment effects and dilution parallelism; use one method consistently within a study.

M3 Net extracellular peroxide accumulation from roots

Basis detection chemistry from M1 and biological context from Liskay et al. (2004). This is a proposed assay for excised roots. It measures net detectable accumulation in the bathing medium, affected by production, release, scavenging, wounding and probe chemistry. It is not a direct measurement of gross apoplastic production or a plant-specific flux in non-axenic roots.

1. Choose and record root class, distance from the tip, segment length, total fresh mass and, where practical, surface area. Rinse in 0.5 mM CaCl_2 plus 5 mM MES, pH 6.0. Start with 30 min equilibration but test the wound-response time course; rinsing cannot be assumed to eliminate it.
2. Incubate standardized segments in 1.00 mL rinse medium containing 50 μM Amplex Red and 0.1 U mL^{-1} HRP in compatible 24-well plates, protected from light. Pilot the detection chemistry at pH 6.0. Keep oxygenation, temperature and agitation fixed; an aerated bath does not reproduce the hypoxic root environment.
3. Use separate matched incubation wells for 0, 15, 30 and 60 min during validation, including catalase-treated roots and root-free controls. A starting catalase concentration is 100 U mL^{-1} , subject to a spike-depletion test. Record the mass in each incubation well.
4. Collect 100 μL medium and read immediately with a consistent short delay. Do not store earlier time points until the end unless a validated stopping procedure and product stability have been demonstrated. Run H_2O_2 calibrators and resorufin recovery checks in the same medium; use no-probe root controls where background is appreciable.

For independent wells, calculate nmol in the full incubation volume at each time point, normalize to that well's root mass, then fit the slope of nmol g^{-1} versus hours over a supported linear interval. For repeated sampling without volume replacement, cumulative product = current concentration $\times$ remaining volume + $\Sigma(\text{previous concentration} \times \text{volume removed})$. Account separately for any replacement medium.

Nonlinearity may reflect biological regulation, scavenging, probe depletion or optical interference. DPI can provide supporting perturbation evidence but is not specific to NADPH oxidases. For 96 samples $\times$ 3 post-zero times $\times$ $\pm$ catalase, 576 incubation wells are needed before zero-time and other controls, equivalent to at least 24 full 24-well plates. Plan M3 separately from the frozen-extract workflow.

M4 Interference corrected TBARS

Basis Hodges et al. (1999) with the equation correction of Landi (2017). Report MDA equivalents or TBARS, not chemically specific MDA. The correction reduces known interferences but does not eliminate all non-MDA products.

1. Extract 100 mg frozen powder in 1.00 mL 80% ethanol. Centrifuge 10 min at $16,000 \times g$ at 4 °C. Prepare +TBA reagent containing 20% TCA, 0.01% BHT and 0.65% TBA; prepare an identical –TBA reagent without TBA. Record how BHT is dissolved and match its solvent in both reagents. Verify complete dissolution.
2. In compatible screw-cap tubes mix 200 µL extract with 200 µL +TBA reagent, and separately 200 µL extract with 200 µL –TBA reagent. This preserves the original 1:1 ratio while avoiding the need to recover a full 1 mL supernatant. The 400 µL reactions do not belong in ordinary 0.2 mL PCR wells.
3. Heat at 95 °C for 25 min in rated closed vessels, cool on ice for 5 min and clarify by centrifugation. Transfer 200 µL of each reaction to a compatible clear reading plate. Read A440, A532 and A600.
4. Run an authenticated MDA calibrator, or TEP-derived MDA with validated complete hydrolysis and concentration, through both reaction branches and heating. Use at least a blank and five nonzero concentrations spanning the samples. Label the x-axis explicitly as nmol mL⁻¹ in the final heated mixture.

Subtract the corresponding +TBA or –TBA reagent blank at each wavelength first. Then $A = (A532 - A600) + TBA$; $B = 0.0571 \times (A440 - A600) + TBA$; corrected signal $S = A - B$. Apply the same calculation to standards. The Landi correction is retained: the extinction coefficient divides the whole $(A - B)$.

Preferred: obtain C_{mix} in nmol mL⁻¹ from the processed standard curve. Content (nmol g⁻¹ FW) = $C_{mix} \times 2 \times D \times V_{ext}/m$. Alternatively $C_{mix} = S \times 10^6 / (157,000 \times l)$, using a validated path length. The reading transfer volume is not another dilution. If standards instead describe the pre-reaction extract concentration, do not multiply by 2 again.

The reacted mixture contains approximately 40% ethanol, not 80%; this still makes a water-only path correction unreliable. Inspect plate compatibility. Include paired QC reactions and recovery checks. Negative corrected signals should trigger investigation and an LOQ flag, not silent truncation to zero. Heating and standard preparation must include controls, so 192 sample reactions do not fit a two-plate heating plan with no spare wells.

M5 Soluble protein by Bradford

Basis Bradford (1976). Protein is required from the same extract used in M6–M9, not from a separate tissue aliquot extracted on another day. Results are BSA-equivalent soluble protein.

1. Choose a specified ready-to-use Bradford reagent and record manufacturer, catalogue and lot. The following 5:250 μL ratio is a starting microplate format; if the selected product specifies a different ratio or incubation, follow that complete product method consistently.
2. Prepare BSA standards at 0–1.0 mg mL^{-1} in the same extraction buffer. Use a separate curve for APX buffer containing ascorbate. Include the same sample dilution solvent and detergent concentration.
3. Add 5 μL standard or diluted extract and 250 μL reagent to each well. Mix, incubate 5–10 min at room temperature and read A595 at a fixed time within the product's validated window. Use duplicates.
4. Fit the validated calibration range and multiply by the extract dilution D. Check at least two sample dilutions and protein spike recovery during validation.

Matching buffers does not guarantee that detergent interference is removed. Check compatibility at the final concentration; dilute into matched buffer or change to a validated compatible protein method if necessary. Keep sample volume constant and do not force a straight line across a visibly nonlinear Bradford curve.

M6 Superoxide dismutase by WST 1 inhibition

Basis Peskin and Winterbourn (2000). Use a named commercial WST SOD kit for a fully specified routine assay, or validate the following proposed 200 μL implementation. It is not claimed to reproduce the original paper's exact formulation. SOD inhibits superoxide-dependent WST-1 reduction; matrix compounds may also scavenge superoxide or inhibit xanthine oxidase.

Pilot final composition: 50 mM potassium phosphate pH 7.8, 0.1 mM DTPA, 0.1 mM hypoxanthine and 50 μM WST-1. Add 20 μL extract dilution, 160 μL concentrated reagent mix and 20 μL xanthine oxidase starter. Prepare reagent concentrations to achieve these final values; titrate the oxidase lot to a measurable, linear uninhibited rate. Record its selected final activity in mU mL^{-1} before routine use; a generic unit range is not a locked method.

1. Prepare three extract dilutions, initially 1:5, 1:20 and 1:50. Use the same final buffer matrix for standards and controls.
2. Set up each dilution with oxidase and a matched no-oxidase well. Include uninhibited reactions with buffer replacing extract, reagent blanks and a SOD standard series.
3. Start in timed blocks, mix and record A450 over 5–20 min at a fixed temperature, initially 25 °C. Choose a validated linear interval. Use an endpoint only after verifying comparable reaction timing and linear accumulation.
4. Correct the sample slope for its own no-oxidase slope. Correct the uninhibited control in the same way.
Inhibition (%) = $100 \times (\text{rcontrol} - \text{rsample})/\text{rcontrol}$.

Interpolate only over a dilution range bracketing 50% inhibition; roughly 20–80% is a useful starting working range. Define one WST assay unit as the amount giving 50% inhibition under the stated conditions. If a dilution factor D50 gives 50% inhibition using 0.020 mL, activity = $\text{D50}/0.020 \text{ U mL}^{-1}$ original extract. Divide by protein in the original extract for U mg^{-1} , or multiply by Vext/m for U g^{-1} . This operational unit is not automatically equivalent to a supplier unit defined using another assay.

No-oxidase blanks detect direct dye reduction but not all interference with the superoxide-generating system. Assess dilution consistency and SOD spike recovery; confirm questionable results by an independent activity assay. Six sample wells are required for three dilutions with paired blanks, before technical duplication.

M7 Catalase activity

Route A UV initial rate

Basis Aebi (1984), with a proposed 200 μL plate adaptation. Use the general extract without added ascorbate or thiol reductants. The assay records apparent H_2O_2 decomposition; check nonenzymatic and other enzymatic peroxide consumption in representative matrices.

1. Prepare 50 mM potassium phosphate pH 7.0 containing 10 mM H_2O_2 , fresh and standardized. Equilibrate substrate and UV plate to 25 °C. With 190 μL substrate plus 10 μL extract, final H_2O_2 is 9.5 mM, not 10 mM.
2. Add 190 μL substrate and initiate with 10 μL diluted extract in a small timed block. Initially use 8–16 reaction wells plus controls; expand only after verifying the actual read cycle and mixing delay.
3. Read A240 every 5–10 s for up to 60 s. Use the earliest valid linear interval after mixing. Dilute extracts until substrate loss is preferably $\leq 10\%$ within the fit interval and the slope is proportional to extract volume.
4. Include extract without H_2O_2 , substrate without extract and buffer-only controls. Inspect bubbles, total absorbance and baseline drift. Use the reader's validated photometric range rather than a universal A = 1 cutoff.

Let s_{EH} , s_{E0} , s_{0H} and s_{00} be signed dA_{240}/dt slopes with extract plus peroxide, extract only, peroxide only and buffer only. Positive net disappearance $R = -[(s_{\text{EH}} - s_{\text{E0}}) - (s_{\text{0H}} - s_{\text{00}})]$. Activity ($\mu\text{mol H}_2\text{O}_2 \text{ min}^{-1} \text{ mL}^{-1}$ original extract) = $R \times 0.200 \times D / (0.0436 \times l \times 0.010)$. $\epsilon = 0.0436 \text{ mM}^{-1} \text{ cm}^{-1}$. Divide by original-extract protein for specific activity; multiply by Vext/m for activity per g FW.

Route B Cobalt endpoint alternative

Hadwan (2018) uses cobalt–bicarbonate as both the reaction-stopping and colour-forming reagent, with detection at 440 nm. It was tested in bacterial and animal materials; maize and wheat plate performance is not established by that paper. The original v1.1 route omitted reaction volumes and reagent concentrations and is therefore not a standalone executable method.

Reference-only option: obtain and implement the complete published recipe and Table 5, including its reagent addition order, 2 min reaction, colour development and blanks, before scaling. Validate plate conversion against a calibrated cuvette method. No unspecified separate stop solution should be added. Use Route A here until this alternative is established locally.

The source uses a first-order calculation. $k = \ln(S_0/S_t)/t$ is a rate constant and is not itself $\mu\text{mol min}^{-1} \text{ mL}^{-1}$ extract. A calibrated endpoint mass balance gives average disappearance activity = $(n_0 - n_t)/(t \times V_a) \times D$, after blank correction; it is not an initial rate when substrate consumption is substantial. The previous recommendation to accept 20–60% consumption for an initial-rate equivalent has been removed.

M8 Ascorbate peroxidase activity

Basis Nakano and Asada (1981). Use the separate ascorbate-containing extract described in A3. The readout is H₂O₂-dependent net ascorbate disappearance at 290 nm; it is an operational APX activity in crude extracts.

Use 200 µL per well: 180 µL premix, 10 µL diluted extract and 10 µL 2 mM H₂O₂ starter. Target final 50 mM phosphate pH 7.0, 0.1 mM EDTA, 0.5 mM ascorbate and 0.1 mM H₂O₂. Dilute APX extracts in the same buffer containing 2 mM ascorbate, so carryover remains constant. With 2 mM ascorbate in the 10 µL aliquot, it contributes 0.1 mM final; the 180 µL premix therefore needs 0.4444 mM ascorbate to supply the remaining 0.4 mM. Account similarly for phosphate and EDTA carried by each component.

1. Equilibrate reaction reagents and UV plate to 25 °C. Set up paired wells for each extract with and without H₂O₂; the latter receives 10 µL matching starter buffer. Include both no-extract reagent controls.
2. Record a short pre-start baseline, then start in timed blocks. Read A₂₉₀ approximately every 10 s for 1–2 min. Fit the earliest linear interval with adequate points and limited substrate depletion.
3. Use the post-addition slopes of the paired wells for correction; the pre-addition baseline is an additional diagnostic, not a substitute for a matched post-dilution blank.

Using signed slopes, $R = -[(s_{EH} - s_{E0}) - (s_{0H} - s_{00})]$. Activity ($\mu\text{mol ascorbate min}^{-1} \text{ mL}^{-1}$ original extract) = $R \times 0.200 \times D / (2.8 \times l \times 0.010)$. Normalize with protein from this same APX extract. Do not take an absolute value to conceal a wrong-sign corrected slope.

The catalytic cycle uses two one-electron ascorbate donations per H₂O₂, but monodehydroascorbate disproportionation and recycling affect net ascorbate disappearance in crude extracts. Do not infer peroxide removal simply by halving the A₂₉₀ activity. APX isoforms vary in substrate-dependent stability; a universal 2 h shelf life is not justified. Test fresh versus delayed assays and work promptly. Change ascorbate concentration only after checking its effect on activity.

M9 Glutathione reductase activity

Route A NADPH disappearance

Basis Foyer and Halliwell (1976). GR reduces one GSSG to two GSH while consuming one NADPH. This measures enzyme activity, distinct from the glutathione pool measured using added GR in M10.

Use 200 μL : 170 μL buffer/NADPH premix, 10 μL diluted general extract and 20 μL 5 mM GSSG. Target final 100 mM phosphate pH 7.5, 1 mM EDTA, 0.2 mM NADPH and 0.5 mM GSSG; account for buffer carried by the extract and starter. If other components contain no NADPH, the 170 μL premix contains 0.2353 mM NADPH. Prepare fresh and protect from light.

1. Equilibrate to 25 °C. Set up paired wells with and without GSSG; replace the starter with matching buffer in the latter. Include both no-extract controls.
2. Record a baseline, start in timed blocks, and read A340 every 10–15 s for 2–3 min. Restrict the fit to an initial linear interval with preferably $\leq 10\%$ NADPH depletion.
3. Confirm proportionality to extract amount. Lower NADPH only after confirming that the chosen concentration supports the intended activity measurement.

$R = -[(s_{EG} - s_{E0}) - (s_{0G} - s_{00})]$, with signed dA_{340}/dt slopes. Activity ($\mu\text{mol NADPH min}^{-1} \text{ mL}^{-1}$ original extract) = $R \times 0.200 \times D / (6.22 \times l \times 0.010)$. Divide by original-extract protein or multiply by V_{ext}/m . At $l = 0.625 \text{ cm}$, 0.2 mM NADPH contributes approximately $A_{340} = 0.778$ before matrix absorbance.

Route B DTNB coupled alternative

Smith et al. (1988) describe a visible GR assay at 412 nm. This route remains reference-only: select the complete published reagent formulation and blank scheme before use. Do not substitute the glutathione-pool assay in M10, which adds exogenous GR and measures a different quantity.

A nominal two-TNB-per-NADPH conversion applies only when the coupled reaction stoichiometry, DTNB availability and steady-state behavior have been established. Endogenous thiols and GSSG regeneration complicate simple background subtraction. The v1.1 formula has therefore been removed as an unconditional recipe. Validate this route against A340 before adopting it.

M10 Glutathione pool and oxidation state

Basis Queval and Noctor (2007), with the explicit 200 μL implementation reported by Stushnoff et al. (2010), doi:10.1093/jxb/erp394. Added GR recycles glutathione; the rate of TNB formation is calibrated against glutathione amount. A timed endpoint can be calibrated in principle, but this SOP uses kinetic rates for robustness.

Extraction and assay

1. Extract 50–100 mg frozen powder in 1.00 mL ice-cold 0.2 M HCl. Centrifuge 10 min at $16,000 \times g$ at 4°C . Keep the acid extract cold and record elapsed time.
2. As a starting neutralization, combine an aliquot with 0.1 volume 0.2 M phosphate pH 5.6 and 0.8 volume 0.2 M NaOH. Check the resulting pH, aiming for 5–6, and adjust if necessary. Record the actual dilution; the stated mixture gives $D_{\text{neutral}} = 1.9$. The final assay must reach its buffered pH near 7.5.
3. Split the neutralized extract. For GSSG, add 1 μL 2-vinylpyridine to 200 μL extract and incubate 30 min at room temperature as a starting derivatization condition. Treat GSSG standards identically. Include a GSH-only control and a known GSH/GSSG mixture. Demonstrate that residual GSH cannot materially explain the measured GSSG; extend or change masking only through revalidation.
4. For each 200 μL assay combine 100 μL 0.2 M sodium phosphate pH 7.5 containing 10 mM EDTA, 10 μL 10 mM NADPH, 10 μL 12 mM DTNB, 10 μL extract and 60 μL water. Initiate with 10 μL GR at 20 U mL^{-1} . Final nominal concentrations are 0.5 mM NADPH, 0.6 mM DTNB and 1 U mL^{-1} GR; buffer from the GR stock also contributes.
5. Read A412 every 15–30 s for up to 5 min. Use a common validated linear interval after any mixing lag. Subtract the matched reagent slope. Prepare separate GSH and processed GSSG curves spanning the samples on each plate.

This explicit implementation does not include an unspecified 2-VP-removal step. If residual reagent suppresses the assay, dilute with matched standards or adopt a documented alternative; do not improvise removal. Standard matching controls reagent effects, but a GSH-only masking control is still essential to demonstrate selectivity. Work in a fume hood with 2-VP.

Calculation and reporting

Label both calibration axes as the concentration in the aliquot solution immediately before its 10 μL addition to the well. Then no 200/10 factor is applied to concentrations read from those curves. Total glutathione T uses a GSH curve and is in GSH equivalents. Oxidized glutathione G uses a GSSG curve and is in molar GSSG.

Convert each branch to nmol g^{-1} FW before subtraction: $\text{content} = \text{Caliquot } (\mu\text{M}) \times \text{Vext (mL)} \times D_{\text{neutral}} \times D_{\text{additional}} \times D_{\text{mask/m}}$. $D_{\text{mask}} = 201/200$ for the GSSG branch above, and 1 for an untreated total branch. If the standards are instead labelled by pre-masking concentration, masking dilution is already included and must not be applied again.

$\text{GSH} = T - 2G$; reduced fraction (%) = $100 \times \text{GSH}/T$; molar GSH/GSSG = GSH/G . Example: $T = 900$ and $G = 100 \text{ nmol g}^{-1}$ gives $\text{GSH} = 700$ and 77.8% reduced. A curve that reports GSSG in GSH equivalents requires subtraction once instead. Keep this convention explicit in data files.

Do not report an infinite ratio when GSSG is below LOQ, and do not force negative calculated GSH to zero. Investigate masking, blanks, oxidation during processing and calibration. These are bulk extract pools, not a compartment-specific glutathione redox potential. Establish stability experimentally rather than asserting a universal 1 h limit.

M11 Reduced and total ascorbate

Basis Gillespie and Ainsworth (2007). The ferric–bipyridyl assay detects ferric reduction; other reductants can contribute. Total ascorbate requires DHA reduction followed by complete quenching of excess DTT. The supplied v1.1 omitted these quantities. The following is a proposed pilot implementation, informed by the plate adaptation in Boestfleisch et al. (2014), doi:10.1093/aobpla/plu046, with a deliberately stronger neutralization buffer. It is not a verified transcription of the 2007 protocol.

Pilot recipe

Use a separate extract: homogenize 50 mg frozen powder in 0.50 mL ice-cold 6% TCA, keep on ice for 15 min and centrifuge 20 min at approximately $18,000 \times g$ at 4 °C. Keep supernatant cold and assay promptly. Do not assume that the M10 HCl extract is interchangeable. Prepare fresh ascorbate standards in 6% TCA, initially 0–1 mM, then select a linear range matching the samples.

1. In each reduced and total well combine 10 μ L extract or standard with 10 μ L 1.0 M potassium phosphate pH 7.4. This buffer strength is a proposed modification: measure the actual mixed pH in a scaled matrix test. Aim for a near-neutral reduction step, about pH 6.5–7.5; do not proceed with an acidic mixture.
2. For total ascorbate add 10 μ L 10 mM DTT, incubate 10 min at room temperature, then add 10 μ L 0.5% N-ethylmaleimide (NEM) and incubate 2 min. For reduced ascorbate add 20 μ L water instead of DTT and NEM, with the same elapsed time. Process separate standard series through the two branches.
3. Prepare colour mix from 2 volumes 10% TCA, 1 volume 3% FeCl_3 , 2.5 volumes 43% phosphoric acid and 2 volumes 4% 2,2'-bipyridyl in 70% ethanol. Percentages for solid solutes are w/v. Define phosphoric acid explicitly as 43% w/v for this pilot (43 g pure H_3PO_4 equivalent per 100 mL final solution), not 43% v/v of an unspecified stock. Record the FeCl_3 hydration state and formula used; prepare 3% as anhydrous FeCl_3 equivalent.
4. Add 100 μ L colour mix to each 40 μ L pretreated sample, giving 140 μ L total. Incubate initially 30 min at 37 °C, cool uniformly and read A525. Confirm the colour-development plateau with a time course; 523 nm in the cited adaptation and 525 nm here must each be calibrated at the chosen wavelength.

Validate the entire pilot recipe before routine use: check mixed pH, full recovery of a freshly prepared DHA standard, equal ascorbate recovery through both branches, and absence of colour caused by incompletely quenched DTT. Use reagent blanks, no- FeCl_3 sample backgrounds with volume-matched solvent, matrix spikes and dilution tests. If these fail, adopt a complete verified reference protocol or an independent chromatographic/enzyme-specific method; do not optimize one reagent while assuming the rest remain valid.

Read Cext from a calibration labelled as pre-assay extract concentration. Content ($\text{nmol g}^{-1} \text{FW}$) = Cext (μM) $\times$ Vext (mL) $\times$ D/m. DHA = total – reduced; reduced fraction = $100 \times \text{reduced}/\text{total}$. No extra 140/10 factor is required with this calibration convention. Propagate the uncertainty of both branches; small differences can make DHA poorly quantified.

Ascorbate oxidase can support specificity validation after appropriate pH adjustment, but it is not active in strongly acidic extracts. Furthermore, DHA produced by oxidase is reduced back in the total-ascorbate branch: an oxidase-treated total aliquot is not a zero-ascorbate blank. Do not interpret this assay as a direct cellular redox potential.

M12 Proline by toluene free ninhydrin

Basis Carillo and Gibon (2011), current PROMETHEUS protocol. The assay is an operational proline-equivalent estimate. Ornithine, other imino acids and pigments may contribute; proline accumulation alone does not establish stress tolerance.

1. Extract 50 mg frozen powder with 1.00 mL 40:60 ethanol:water overnight at 4 °C. Centrifuge 5 min at 14,000 × g. Keep the extraction duration fixed. This cold extraction is an explicit option in the source; other extraction schemes must not be mixed within a dataset.
2. Prepare reaction reagent containing 1% ninhydrin in 60% acetic acid and 20% ethanol. In rated screw-cap tubes mix 50 µL extract with 100 µL reagent. Include solvent-matched proline standards, initially 0, 2, 5, 10, 20 and 50 nmol per 50 µL aliquot.
3. Heat at 95 °C for 20 min, cool uniformly, briefly centrifuge, and transfer 100 µL of each 150 µL reaction to the same type of compatible clear plate. Read A520 within a validated fixed window. Process standards identically.

If the calibration x-axis is nmol in the original 50 µL aliquot, the transferred 100/150 fraction is already captured by the curve. Proline ($\mu\text{mol g}^{-1} \text{FW}$) = $n_{\text{aliquot}} \times (V_{\text{ext}}/0.050) \times D/(1,000 \times m)$, with volumes in mL. Example: 10 nmol, 1.00 mL extract and 0.050 g tissue gives 4.0 $\mu\text{mol g}^{-1}$. Do not multiply by reaction volume.

Use reagent and solvent blanks, matrix spikes and dilution checks. For coloured extracts, run matched no-ninhydrin backgrounds and validate that subtraction is adequate; it does not remove all reaction-generated interference. Confirm critical findings chromatographically. Do not transfer the classical toluene extract into polystyrene. Avoid unvalidated sealed PCR plates for hot acidic ethanol reactions.

C Imaging methods and scope limits

Method	Appropriate interpretation
DAB and NBT histochemistry	Preserve the spatial readout with standardized imaging. Bulk solubilization is a different assay requiring independent validation.
H ₂ DCFDA	A loading-, esterase-, oxidation- and efflux-dependent fluorescence index; not a specific H ₂ O ₂ or total ROS concentration. Bulk wells lose spatial information.
CeCl ₃ and EPR	Require appropriate microscopy or spectroscopy; not routine plate-reader substitutions.
roGFP and related sensors	Some sensors can be measured ratiometrically in suitable plate readers. Localization, tissue optics and oxidized/reduced calibration determine validity; they are not universally restricted to microscopy.

D Plate planning and calculations

D1 Capacity for 96 biological samples

For planning, reserve 24 of 96 wells on every plate for standards, controls and QC, leaving 72 sample wells. These are minimum arithmetic estimates, not promises of biological or technical replication. Expand control allocation where required. If the outer ring is excluded, capacity is lower. Use edge wells only after demonstrating acceptable evaporation and spatial uniformity.

Assay	Sample wells per biological sample	Minimum plates or runs
M1	4 with paired catalase and no-probe controls; 8 with duplicates	6 plates; 12 with full duplication
M2	2; 4 with duplicates	3; 6 with duplicates
M3	Separate living-root incubation design	Calculate time × control × replicate count
M4	2; 4 with duplicates	3; 6 with duplicates
M5	2 technical wells per actual extract	3 plates per extraction buffer type
M6	6 for three dilutions and paired blanks	8 plates; 16 with full duplication
M7 A	2 paired sample wells plus shared controls	At least 3 plates by capacity; more timed blocks
M7 B	Reference-only alternative	Not scheduled until validated
M8 and M9 A	2 paired sample wells each	At least 3 plates each; typically smaller kinetic runs
M10 and M11	2 branches; 4 with duplicates	3; 6 with duplicates, plus matrix-specific blanks
M12	1; 2 with duplicates	2; 3 with duplicates; extra pigment blanks if needed

D2 Usable plate layouts

Replace the original full-plate grids with the layouts below. Randomize biological sample identity within each sample block; retain adjacency of paired reactions. Standards and QC need not always occupy the same edge positions once spatial performance is verified.

Assay	Sample block	Control block
M1 full controls	Columns 1–8: each column has two samples. A–D and E–H each contain untreated/probe, untreated/no-probe, catalase/probe and catalase/no-probe. Capacity 16 samples.	Columns 9–12: 7 standard levels × 4 matched conditions = 28 wells, plus pooled QC × 4 = 4 wells. One plate is full; use separate validation plates for spikes.
M4	Columns 1–8: 32 samples, each occupying adjacent +TBA and –TBA wells across column pairs 1/2, 3/4, 5/6 and 7/8.	Columns 9–12: seven standard levels including zero × two branches × duplicates = 28 wells, plus paired QC in duplicate = 4 wells.
M8 and M9 A	Start with 8–16 samples in paired wells, then expand after verifying timing and slope linearity.	Include both substrate-present and substrate-absent no-extract controls and repeated pooled QC. Do not allocate every read well to samples.

The M1 layout fits 16 samples per plate and requires six plates for 96 samples without technical duplication of the four conditions. It is consistent with the 0–5 μ M in-well standard range. The M4 layout fits 32 samples, so three plates accommodate 96 samples. Controls must also be included in the heating step.

D3 Kinetic timing

A multichannel pipette starts one row or column at a time, not an entire plate. Instrument scan order, per-well timestamps, injection capability and dead time are model-specific. A constant start offset need not bias a stable linear slope, but it can miss the initial phase or cause unequal substrate depletion. Test this directly; there is no universal 24–48-well limit.

Use the exported measurement times for each well where available. Confirm the actual cycle duration with the selected wavelengths and wells. Injectors help only if they support the required reagent, volume and read sequence; they do not remove all timing constraints. For stopped endpoints, start and stop each block in the same order with the same elapsed reaction time.

D4 Work allocation

First establish the assays on a small representative set. For the main experiment, harvest into frozen aliquots for each extraction family; retain separate living material for M3. Run M1 in small fresh-extraction batches. Run M4 and M12 separately, allowing overnight proline extraction. On enzyme days, extract only what can be assayed promptly and measure protein in those same extracts; prioritize APX. Run M10 from HCl extracts and M11 from separate TCA extracts unless shared extraction is independently validated.

Do not promise a five-day completion time before timing the pilot. Several hundred sample wells plus repeated calibrations, masking and heating steps require substantial preparation. Use frozen pooled tissue aliquots or a demonstrated stable reference material for between-day QC; a single pooled extract may degrade and cannot automatically serve across weeks.

D5 General equations and worked activity example

Let V_a be the volume of extract added to the reaction, V_r the final reaction volume, V_{ext} the original extraction volume, D the pre-assay extract dilution and m the extracted fresh mass. All volumes below are in mL. Use distinct names for V_a and V_{ext} ; the original document used $V_{extract}$ ambiguously for both.

Quantity	Equation
Positive substrate disappearance	$R = -(\text{blank-corrected signed } dA/dt)$
Concentration disappearance	$v = R/(\epsilon \times l)$, in $\mu\text{mol mL}^{-1} \text{min}^{-1}$ when ϵ is $\text{mM}^{-1} \text{cm}^{-1}$
Original-extract activity	$U \text{ mL}^{-1} = v \times V_r/V_a \times D$
Specific activity	$U \text{ mg}^{-1} \text{ protein} = \text{original-extract activity} / \text{original-extract protein in mg mL}^{-1}$
Fresh-weight activity	$U \text{ g}^{-1} \text{ FW} = \text{original-extract activity} \times V_{ext}/m$
Metabolite content	$\text{nmol g}^{-1} \text{ FW} = C_{ext} \text{ in } \mu\text{M} \times V_{ext}/m$; apply only dilutions not already included in C_{ext}
M10 reduced glutathione	$\text{GSH} = \text{total GSH equivalents} - 2 \times \text{molar GSSG}$
M11 oxidized ascorbate	$\text{DHA} = \text{total ascorbate} - \text{reduced ascorbate}$

GR example: $R = 0.020 \text{ A min}^{-1}$, $\epsilon = 6.22 \text{ mM}^{-1} \text{cm}^{-1}$, calibrated $l = 0.625 \text{ cm}$, $V_r = 0.200 \text{ mL}$, $V_a = 0.010 \text{ mL}$ and $D = 5$. Then $v = 0.005145 \mu\text{mol mL}^{-1} \text{min}^{-1}$ and activity = 0.5145 U mL^{-1} original extract. At $2.0 \text{ mg protein mL}^{-1}$, specific activity = 0.2572 U mg^{-1} . With $V_{ext} = 1.00 \text{ mL}$ and $m = 0.100 \text{ g}$, activity = $5.145 \text{ U g}^{-1} \text{ FW}$.

D6 Analysis and records

Average accepted technical replicates within each biological sample after QC. Retain the original well signals, calibration fit, timestamps, blank corrections, dilution factors, tissue mass, buffer recipe, extraction interval, reagent lots, plate type, reader settings and fit window. Flag failed or below-LOQ results explicitly.

Model the biological design, including treatment factors, interactions and repeated sampling where relevant. Account for plate or extraction batch if supported by the design; with only a few plates, a fixed batch term

may be more appropriate than estimating a random-effect variance. No statistical adjustment can recover a treatment effect perfectly confounded with plate. Do not normalize away plate means when those means may contain real treatment differences.

E Evidence and remaining validation

The review checked the supplied document against accessible original methods and manufacturer instructions. The most direct procedural checks were the Brumbarova Amplex Red method, Junglee KI method, Hadwan catalase method, Landi TBARS correction, the published glutathione implementation in Stushnoff et al., and the Carillo–Gibon proline protocol. Some foundational references were available only through bibliographic records or later primary applications; exact replication is not claimed for every recipe.

Before releasing this as a laboratory SOP, record the instrument model, compatible plates, selected reagent products, approved extraction recipes, standard preparation records, stability intervals and pilot QC results. Lock the SOD oxidase activity and the ascorbate pilot conditions. If the colorimetric CAT or DTNB GR alternatives are selected, attach their complete validated procedures. These are concrete unresolved method choices, not evidence that the corrected calculations are uncertain.

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